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# PROCEEDINGS

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## **Cost Estimating in Construction Projects: The Relationship Between Cost and Macro Economic Indicators**

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### **Abstract**

*Accuracy of cost estimating in construction projects is one of the most fundamental issues for each of the stakeholders involved in projects in the construction sector. One of the main approaches to cost estimation is based on the use of macroeconomic indicators. In this study, an approach to integrate the use of indicators related to macroeconomics with artificial intelligence in cost estimating and to create the infrastructure of cost estimating tools that are as strong theoretically as practically is examined. For this purpose, the official square meter cost data of construction projects of reinforced concrete buildings between the years 2010-2020 in Turkey have been used to investigate the above stated relationship between economic indicators and cost of the projects, to demonstrate the relationship between them. Consequently, it has been concluded that economic indicators can be used in the determination of costs of construction projects. Furthermore, this approach provides a more effective cost estimating method compared with the other approaches/models proposed on this subject since it allows to determine much more relationships between economic indicators and cost with the support of artificial intelligence tools in practical dimension.*

**Keywords:** *Construction projects, cost estimating, economic indicators, reinforced concrete buildings, cost per square meter, artificial intelligence.*

### **Introduction**

The cost estimate of projects closest to the actual cost is one of the main challenges that stakeholders of construction projects face. While Bledsoe (1992) underlines that there are studies on cost estimates to be adopted at various stages of projects and intended to prove many arguments/hypotheses on accuracy/precision, Zakwan and Ara (2011) argue that there are lots of tools and methods for cost estimation.

Scientific studies on macroeconomic indicators in construction business analyze the dynamics of the business (Akintoye & Hardcastle, 1998). Macroeconomic indicators are one of the methodologies to estimate construction costs, and there are many scientific studies on them.

Among them are studies by Oteng-Abayie and Dramani (2018), Arıoğlu and Girgin (2003), Olga and Antonios (2019) and Diler (2018). While the aforementioned studies are focused on the correlation between any situation in construction business and macroeconomic indicators in their context, they do not address how macroeconomic indicators can be utilized as a cost estimation instrument. In addition, the aforementioned studies are limited in context, and as they do not consider all the factors that affect costs of construction projects, they do not offer any practice and means to make inferences, and any solution on how to effectively/reliably update cost estimation data as it remains a major problem.

The large-scale and complicated nature of modern construction projects and constant and instantaneous changes in today's globalized world leave no choice but to continuously revise original cost estimates of construction projects at a short notice, and make the revisability of cost estimates a major problem.

**Objective:** The aim of this study is not only to reveal the correlation between construction costs and foreign exchange rates as a well-known case but also to set forth how to correlate cost items/cost components, which are generated based on a core analysis of items of construction costs, with economic indicators based on a sample. This study will establish a cost estimate and revision model depending on macroeconomic indicators irrespective of cost estimation methods. This will help outline a paradigmatic and conceptual model of cost estimation that can be configured by artificial intelligence to estimate and revise construction costs based on economic indicators. Relying on the establishment of a correlation between project cost estimates and continuously monitorable/accessible economic indicators, this revisable cost estimation model will provide a major input to minimize deviations from original cost estimates. This study is an attempt to combine the use of macroeconomic indicators and artificial intelligence for cost estimation, and address an approach that can serve as a basis for the infrastructure of cost estimation instruments that are theoretically and practically favorable to use. Key concepts, opportunities and barriers of this approach intended to suggest such an input are going to be addressed, and a proposed model is going to be outlined as a part of the study.

**Scope:** For the scope of the study, a data set that enables to make a significant and weighted/high-rate correlation with macroeconomic indicators generated as a result of country-wide complicated and dynamic relations. In this context, the scope of the study was designated as residential buildings constructed based on a reinforced concrete load-bearing system, and this is based on the fact that residential buildings rank first among all types of buildings in Turkey, making up 84.62% of them, and 99% of the residential buildings are equipped with a reinforced concrete frame/reinforced concrete load-bearing system (TURKSTAT, 2020). The study will attempt to achieve consistency of a chain ranging from a macroeconomic scale to the extreme/core cost within the boundaries of the aforementioned limitations. This research is limited to investigating the relationship between the economic indicator of foreign currency (USD) and the cost per square meter of housing. However, it should not be forgotten that it is open to expandable research on cost relations with other economic indicators.

**Method:** An economic indicator-construction cost relation will be mapped out based on weight/rate of macroeconomic building cost estimates to be addressed as a part of the study to establish a significant and consistent correlation between macroeconomic indicators, which are outputs of complicated correlations at a macroeconomic level, and construction costs. In other words, a chain of relations will be established based on a weighted/high-rate data set. This will show that economic indicators can serve to make construction cost estimates, and an effective

cost estimation method can be introduced with a lot more correlations between economic indicators and costs based on means of artificial intelligence. A sample structure was created for learning with the *decision tree*, which is one of the artificial intelligence methods, using statistical distribution, arithmetic mean and correlation methods.

**Data Set:** The data of the Turkish Statistical Institute (TÜİK) and publicly-available data of the Turkish Ministry of Environment and Urbanization, the Presidency of Strategy and Budget, the Turkish Central Bank, and trade unions from 2010 to 2020 are going to be utilized for the study. Details on sources of data and calculations are as set out in TURKSTAT's reference statement on data (TURKSTAT, 2020a). The data of the study are data released in official documents and serve as a basis for major legal procedures such as taxation (Turkish Official Gazette, 1970). Testing the reliability of the data is beyond the scope and purpose of this study. The structure of the primary data set used in the study and the methods used in its creation are explained in TURKSTAT resources. On the other hand, it is not possible to access primary sources of secondary data, some primary data have been accessed for the purpose of verifying secondary data; However, disclosure of these data according to The Turkish Penal Code (TPC) no.5237 embodies "the reveal of confidential business information and documents" as a crime could not be presented because it is a commercial secret.

Economic indicators, which are a key part of the data set to be utilized as a part of the study, are statistics about economic operations and data that enables to analyze those operations and make prospective estimations. In addition, economic indicators have the potential to make significant inferences as they are generated by a statistical analysis of many data. Such statistical data are collected by the Organization for Economic Co-Operation and Development (OECD), the Organization for European Economic Co-operation (OEEC) and many statistical institutes from a great deal of sources. The main economic indicators, which are established by TURKSTAT in Turkey, are categorized by the TURKSTAT, and the Presidency of Strategy and Budget based on national income, production, fixed capital investments, equilibrium of foreign trade and payments, money, bank, capital market, prices, employment and fees.

In this study, it has been concluded that the building type is residential and reinforced concrete structural system is used in residential buildings, with statistical distribution based on TURKSTAT construction m2 cost data between 2010 and 2020. Reinforced concrete structural system costs, data on 68 housing projects were analyzed and it was concluded that the statistical distribution, steel and concrete costs were among the inputs that significantly affect the building cost. When the inputs that constitute the cost in the production of steel and concrete with the statistical distribution were examined, it was seen that the common and highest cost input was the energy cost. When the acquisition of energy is investigated, it is concluded that foreign dependency in energy is carried out with foreign currency (USD). In order to determine what the root input cost of the construction m2 cost is dependent on foreign currency (USD), by selecting the 'luxury housing m2 cost' data, which includes more imported inputs, from the average annual exchange rate values between 2010-2020 and the housing cost data between 2010-2020. The correlation between exchange rate-m2 housing cost has been calculated. In order to question the continuity of the dependency relationship between the exchange rate and the cost of housing, a comparison was made with the exchange rate housing m2 cost data between 1986-1993 for a reinforced concrete house. In conclusion, it has been revealed that the strong relationship between the exchange rate and housing m2 cost with the correlation values in the graphical representations of 2010-2020 existed in the past and continues today, and a cost estimation model based on this strong relationship can be suggested.

## Establishment of a Correlation between Macroeconomic Indicators and Items of Cost

Main cost items are composed of production inputs as it is well known. Once a cost correlation chain is formed based on items of production with the highest rate of cost, it is possible to draw up a cost correlation map that typifies the cost. To define this cost correlation map based on a sample of residential buildings constructed by a reinforced concrete load-bearing system designated as the scope of the study, the main production items of cost and economic indicators likely to be correlated were analyzed, and it led to the following results:

**Residential Construction:** Residential buildings make up 80.35% to 87.08% of buildings constructed in Turkey according to the figures of the TURKSTAT on New and Additional Buildings Constructed for Intended Use Based on Building Permits Issued from 2002 to 2020. The mean value of 19-year data shows that residential buildings make up **84.62%** of all types of buildings, ranking first (TURKSTAT, 2020b).

**Reinforced Concrete Frame Buildings in Residential Construction:** The data of the TURKSTAT on Intended Use and Load-Bearing System by the Type of Building Permit from 2015 to 2020 were collected to analyze residential buildings in Turkey. The figures show that 97% of all buildings are of reinforced concrete frame, and the rate goes up to **99%** in residential buildings.

**Reinforced Concrete Load-Bearing System:** Systems that are commonly known as reinforced concrete skeleton structural systems are load-bearing systems where static and dynamic loads are carried by structural elements such as floor cover, columns, beams and foundation, and they are shifted onto the foundation. As it is known, a building cost is composed of many items of cost. The literature review shows that the cost of a reinforced concrete load-bearing system is much higher than the cost of other items. Seyfi et al. (2017) reported that the cost rate of a load-bearing system for residences of 4 to 8 floors is **34.52%** on average as presented in Table 1. Esra et al. (2004) compared costs of masonry, wooden and reinforced concrete load-bearing systems and reported that the cost rate of the reinforced concrete load-bearing system is 26%. In a way to corroborate the aforementioned results, Saner (1993) reported that the cost ratio of reinforced concrete load-bearing systems to other costs is over **40%** as in Pişirici (1981) research

Table 1. Analysis of main items of cost (Seyfi et al., 2017).

| Main Cost      | Min. Cost            | Avrg. Cost           | Max. Cost            | Min. Percentage | Average. Percentage | Max. Percentage |
|----------------|----------------------|----------------------|----------------------|-----------------|---------------------|-----------------|
| Items          | (TL/m <sup>2</sup> ) | (TL/m <sup>2</sup> ) | (TL/m <sup>2</sup> ) | (%)             | (%)                 | (%)             |
| Rough Works    | 219,323              | 372,688              | 516,603              | 24,28%          | <b>34,52%</b>       | 49,54%          |
| Finishings     | 317,036              | 472,685              | 778,635              | 33,30%          | 43,20%              | 54,44%          |
| Electr. Wiring | 24,245               | 61,621               | 125,014              | 2,87%           | 5,53%               | 8,64%           |
| Gas Inst.      | 20,550               | 63,387               | 134,295              | 2,19%           | 5,74%               | 8,40%           |
| Plumbing       | 35,076               | 73,165               | 136,452              | 3,60%           | 6,60%               | 10,00%          |
| Elevator       | 15,428               | 50,379               | 105,56               | 1,86%           | 4,42%               | 7,69%           |
| Total          | -                    | 1.093,93             | -                    | -               | 100%                | -               |

The figures revealed that it is possible to make a projection about the total construction cost based on the cost of a reinforced concrete load-bearing system, which has a large share in total

cost. In addition, it was considered that simple and computable items of cost such as concrete, steel and mould in this sample will provide an advantage to establish a clear correlation between economic indicators and items of cost.

**Items of Cost for A Reinforced Concrete Load-Bearing System:** Ready-mixed concrete, ribbed steel, hollow bricks, wooden moulds and supporting scaffolds were regarded as items of construction costs with a reference by the Ministry of Environment and Urbanization in a dissertation titled "Estimating the Cost of a Reinforced Concrete Load-Bearing System Based on Artificial Neural Networks" by Ünsal (2017) who analyzed the cost of reinforced concrete load-bearing systems. Performed based on the cost estimates of reinforced concrete load-bearing systems in 68 actual projects included in the dissertation, the study revealed the following results that are not indicated in the aforementioned dissertation:

- It was reported that 54 out of 68 projects (79.41%) were built on C30/37 ready-mixed concrete, Ø8- Ø12 mm ribbed steel and Ø14- Ø32 mm ribbed steel.
- Of the total cost computed based on all items of cost, their unit prices and quantities, it was reported that **ribbed steel made up 31.71% of the total cost, and the ready-mixed concrete made up 22.98% of the total cost**, and both combined made up **54.69%** of the total cost. These results show that the cost of ribbed steel and ready-mixed concrete, two items of cost for a reinforced concrete load-bearing system, is higher than the cost of all the other items of cost such as moulds, scaffolds and hollow bricks. Performed to establish a correlation between the core cost and economic indicators, the analysis on the cost of ribbed steel and ready-mixed concrete led to the following results:

**Cost of Steel:** As for items of cost in steel production, steel as a raw material ranks first and electricity energy cost ranks second making up 15 to 25% of the total cost. It is reported that more than 50% of primary raw materials needed for Energy Efficiency Works in Iron & Steel Industry are imported (Turkish Ministry of Development, 2014) This shows a direct correlation with foreign exchange rates as steel as a raw material and electricity rely on importation.

**Cost of Ready-Mixed Concrete:** "Aggregate, which makes up 75% of ready-mixed concrete, is produced naturally from coasts and river beds or artificially from stone crushers. The cost of energy spent on machines operated to produce aggregate and use for concrete mix, and of the energy spent for shipment is equal to the total energy cost of concrete production. However, it is the energy spent for cement production that takes the largest share of energy consumed for concrete production. The energy spent for cement production is equal to over **90%** of the energy spent for concrete production (TUBITAK MAM, 2018). That is why cement costs were analyzed.

**Cost of Cement:** Items of cost for cement production, which is one of the items of production cost for reinforced concrete load-bearing systems, are other extreme-core items of cost, and items of cost that enables to make a correlation for economic indicators. An analysis on the items of cost for cement production shows that fuel and energy costs made up 33% to 40.7% of the total cost in cement production described as dry and wet systems from 1964 to 1968, having the largest share of cost compared to others (MPM, 1970). The Report of the Ministry of Science, Industry and Technology on Cement Industry (2013) indicates that fuel and electricity costs have the largest share in total cost, and *fuel* costs currently make up **38%** of the total cost while *electricity* costs make up **21%** of the total cost. In addition, raw materials and auxiliary materials makes up 9.60%, Laborship is 9.40%, amortization is 7.00% and other expenses are 14.90%. The figures show that fuel and electricity costs in total, which are energy costs in other

words, amount to 59.10% of the total cost. Despite saving on fuel and energy over time thanks to cutting-edge technologies that serve for production, it is still the top item of cost for energy, and likely to remain so in the future. These results are repeated in a verifiable way and detailed in terms of the types of fuel used for energy in Çevik (2016) research. These results show that energy input cost is the main item of cost that is decisive for the cost of cement production. One can infer from this result that the cost reinforced cement namely the cost of reinforced cement construction has a clear and significant correlation with the cost of energy.

**Energy Cost:** As for how energy is supplied, Turkey seems to be forced to keep importing energy as it meets more than 75% of its energy needs from imported resources (Özalp, 2019). Reliance on import brings about foreign currency as a means of purchase. Apart from imported resources, the unit price of purchase for energy to be generated in Turkey under the Law No. 5346 of 5/15/2005 on Utilization of Renewable Energy Resources for the Purpose of Generating Electrical Energy is also based on USD cent. In addition, energy prices tend to be based on USD as it is acknowledged as a reserve currency in the global economy. As a result, the fact that Turkey meets whopping 75% of its energy needs from imported sources in USD clearly reveals correlation with USD as one of the macroeconomic indicators. Likewise, Bilirgen (2019) reported that electricity, oil and gas prices are set based on foreign exchange rates and supply-demand equilibrium. In an article intended to provide data for energy producers and titled “High exchange rates multiple energy costs”, Oral (2017) reported that the effect of USD as a foreign currency on energy prices is dramatic.

**Economic Indicator:** Foreign currency is an economic indicator that projects many economic-political relations at a global and domestic scale. In addition, *the foreign currency (USD)* is an economic indicator for the cost chain correlation established in the light of the data in the study. The fact a foreign exchange rate is a data to be instantly revised based on economic indicators provides a means to roll out an instantly-revisable cost estimation model based on foreign exchange rates.

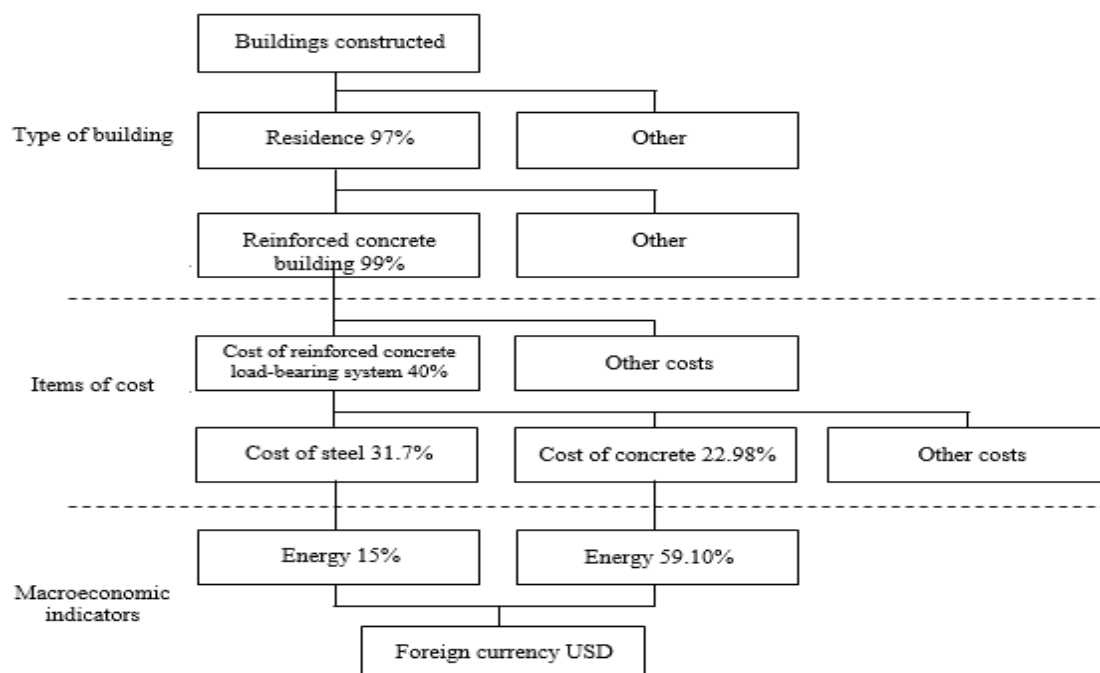


Figure 1: Items of weighted/high-rate construction cost and macroeconomic indicators.

Macro-economic indicators in the levels of literature survey revealed the relationship to the cost of construction, was built between building types in Turkey was seen as having the highest share of residential construction type. In the total cost of housing construction, it has been observed that the cost of the structural system has the largest share, and the cost of the structural system has the highest share in the cost of steel and concrete. It was concluded that the energy cost has the highest share in the steel and concrete cost inputs. Thus, the relationship chain extending from macroeconomic level to cost input level has been reached. The relationship chain expressing the artificial intelligence decision tree model is shown in Figure 1.

The analytical results of building cost estimates point to a correlation between the cost of a building and foreign exchange rate, one of the macroeconomic indicators. Revealing a correlation between the cost of a building and foreign currency, this result shows that economic indicators can serve as a means of cost estimation. Proving the established relation by tangible data; the relationship between m2 costs by years and the exchange rate of a reinforced concrete housing project, which is taken as an example in the thesis study based on the real project data between 1986-1993, is stated. Figure 2: correlation between USD and cost per m2 1986 is very high, such as 0,998902 (99%) (Yazıcıoğlu, 1994).

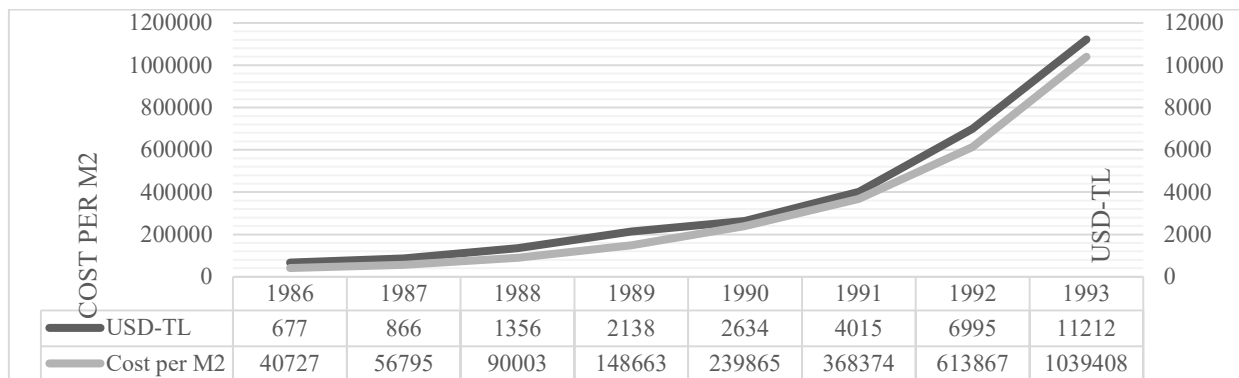


Figure 2: 1986-1993 years USD-Cost per m2.

In this study, based on data between 2010 and 2020, it is stated that the relation between reinforced concrete housing m2 building cost and exchange rate have continued strong and continuously. It is seen that the correlation between USD and cost per m2 between 2010 and 2020 has continued at a very high level such as 0.926683 (92%) Figure 3.

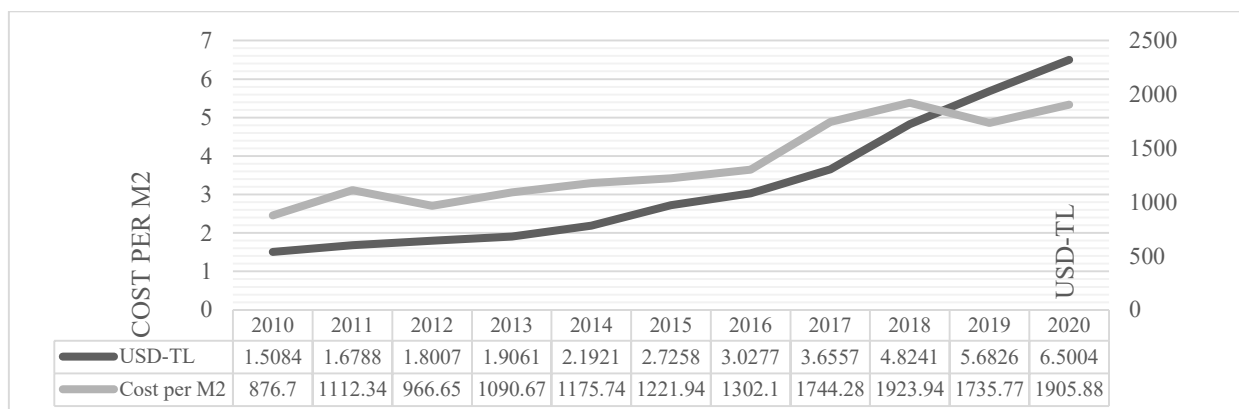


Figure 3: 2010-2020 years USD-Cost per m2.



The results of the analysis of building cost data in both 1986-1993 and 2010-2020 show that the relation between building cost and macroeconomic indicators exists in the past and present. In another study that supports these results, it is stated that there is a bidirectional causality relation between exchange rates and house sales prices, and imported raw materials, which can be a cost element in house sales prices, are affected by the exchange rate of the country (Diler, 2018). All these results evidently demonstrate that the exchange rate can be used from macroeconomic indicators in a model for cost estimation and updating.

## **Conclusion and Recommendations**

The study not only introduces a new approach to cost estimation methods that are already available but also enables to revise cost estimates irrespective of cost estimation methodologies. This will improve the efficacy of cost control and management.

Based on this study, one can argue that more correlations between items of cost and economic indicators and introduction of artificial intelligence would lead to a much more effective and revisable cost estimation model. Each correlation with economic indicators constitutes a part of a cost estimation model that enables to expand its scope, and a cost estimation model would improve in efficacy and precision of results as more and more correlations with economic indicators are established.

As a similar set of rules established for the sample of this study can be utilized as a method to correlate different items of cost with different economic indicators, As Makarov (2019) study on the relationship between building materials and economic indicators points out, other economic indicators and different cost levels, for example; In the case of researching and expanding the effects of economic indicators on construction costs such as oil prices-insulation material cost, unemployment values-labor cost, it will be possible to create a cost estimation model for multiple economic indicators-cost relationship in cost estimation by revealing cross-economic indicators and cost relations.

The results obtained from the weighted distribution of the data in the research from the statistical methods exemplify the possibility of constructing a configurable artificial intelligence model between economic indicators and construction costs with the "learning with decision tree" methodology, which is one of the artificial intelligence methods. These results obtained from the limited data set sampled in the research can be expanded to other economic indicators and cost relationships at different levels (such as oil prices-insulation material cost, oil prices-excavation costs, unemployment values-labor cost) through data mining methods. It will provide the opportunity to construct more advanced decision tree learning models over the new data sets created by the expansion of the data set. Integrated models in which decision tree models are structured together to produce cross results will increase the efficiency of cost estimation. In particular, online updatable structures to be established over computer networks with instantly updatable data such as the exchange rate in the research sample will provide instant monitoring of costs during the project process. As a result, the project will be able to manage decision-making processes much more effectively.

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